Prabhavit Gupta

AI Engineer | Expert Python, PostgreSQL + pgvector, Multi-Agent Systems & Langfuse

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (EU Permanent Resident | Remote Spain/Madrid | Immediate Availability)

Executive Summary

- **Production AI Engineering & Systems Architecture:** Software engineer with 3 years commercial experience architecting, benchmarking, and shipping scalable AI systems, multi-agent pipelines, and vector retrieval backends. Independent builder with an applied engineering backbone, delivering robust end-to-end features using expert Python, async I/O, Docker, and CI/CD.
- **Vector Databases, Relational Schemas & Ingestion:** Deep expertise in PostgreSQL with pgvector, HNSW indexing, pg_trgm similarity matching, and SQLAlchemy/Alembic data migrations. Experienced designing high-throughput data pipelines that deduplicate and reconcile messy enterprise records against authoritative datasets.
- **Multi-Agent Orchestration, Evals & Observability:** Proven track record implementing multi-agent workflows (separating classification agents from reasoning agents), Chain-of-Thought prompting, and guardrails. Extensively instrumented distributed trace observability with Langfuse and automated regression test harnesses (RAGAS) to maintain explainability, quality, and low latency.

Technical Skills & Core Competencies

- **Core Engineering:** Expert Python (Typing, AsyncIO, Performance Optimization, PyTest), SQLAlchemy, Alembic, Docker, REST & WebSocket APIs, Microservices, Git, GitHub Actions CI/CD.
- **Databases & Vector Search:** PostgreSQL (pgvector, HNSW index optimization, pg_trgm fuzzy text matching, partitioning), Redis (Caching & Semantic Queues), Relational Data Modeling.
- **AI Orchestration & Agents:** Multi-Agent Systems (Classification & Reasoning Agents), Chain-of-Thought Prompting, LangChain, LangGraph, Model Context Protocol (MCP), Context Pruning.
- **LLM Observability & Evals:** Langfuse Distributed Tracing, Latency & Token Cost Profiling, RAGAS Benchmarking, Regression Detection Test Suites, Guardrails (Input/Output Validation).
- **Modern Productivity:** AI-Assisted Engineering Workflows, Rapid Prototyping, Clean Code Architecture, Extreme Independence & Ownership.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and shipped scalable production AI systems and multi-agent workflows for European client platforms, taking requirements from system design to high-throughput cloud deployment.
- Developed specialized multi-agent pipelines separating classification agents from deterministic reasoning agents, automating entity resolution and deduplication against external registers.
- Optimized PostgreSQL database architectures with pgvector (HNSW indexing) and pg_trgm, improving semantic and hybrid similarity search query speeds across millions of vectors.
- Standardized agent-to-tool interfaces using Model Context Protocol (MCP) and SQLAlchemy, enforcing strict Pydantic schemas and defensive validation across database read/write actions.
- Instrumenting comprehensive real-time observability with Langfuse, tracking execution traces, token burn, and error fallbacks to ensure robust, explainable system behavior.
- Established automated evaluation harnesses using RAGAS and regression test suites, catching quality degradation and hallucination drift prior to production releases.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered asynchronous microservices in Python (FastAPI) and TypeScript for automated extraction, semantic parsing, and classification of high-volume digital documents.
- Built semantic caching and vector search layers using PostgreSQL with pgvector and Redis, cutting redundant inference calls and maintaining sub-second query latency.
- Authored exhaustive unit and integration test suites using PyTest, establishing continuous quality gating in GitHub Actions CI/CD workflows.
- Collaborated with cross-functional technical teams to convert complex analytical requirements into modular, maintainable software architectures.

Key Engineering Projects

- **Multi-Agent Entity Attribution & Deduplication Pipeline (Python + LangGraph + pgvector):** Multi-agent collaborative system separating categorization agents from analytical validation agents, cross-referencing

records against public registries with Langfuse tracing.

- **High-Throughput Vector Ingestion & Search Engine (FastAPI + pgvector + HNSW):** Hybrid search service combining semantic embeddings with pg_trgm fuzzy matching, automated Alembic migrations, and automated RAGAS eval gates.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, statistical reasoning, decision theory, and complex systems analysis.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Spanish (Elementary understanding).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the EU/Spain without visa sponsorship; available immediately for remote/Madrid collaboration.